Computer Vision – Filters

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Filters

1. Given the following single row of an image with 9 gray levels (0-8):
   1, 0, 2, 2, 2, 8, 2, 3, 6, 6, 6
   Filter this image row
   a. with filter mask 1/3 [1 1 1]
   b. with filter mask [1 0 -1]
   c. with a 1x3 median filter

2. The following table shows a part of a gray value image. Filter the 4 pixels in the middle (printed in bold) and enter the results in the empty tables below.

1	1	4	4
1	**1**	**4**	4
1	**1**	**4**	4
1	1	4	4

   a. use a 3x3 mean filter

   b. use a 3x3 median

3. The following table shows a part of a gray value image. Filter the 24 pixels in the middle (printed in bold) and enter the results in the empty tables below.

1	1	1	1	4	4	4	4
1	**1**	**1**	**1**	**4**	**4**	**4**	4
1	**1**	**1**	**1**	**4**	**4**	**4**	4
1	**1**	**1**	**1**	**4**	**4**	**4**	4
1	**1**	**1**	**1**	**4**	**4**	**4**	4
1	1	1	1	4	4	4	4

a. Use a 3x3 Sobel (derivation in x-direction)

b. use a 3x3 Sobel (derivation in y-direction)

Python/OpenCV - Task 1

Familiarize yourself with OpenCV. In particular, try out the following functions with a test image of your choice:

- Reading and writing images: `imread`, `imwrite`, `imshow`
- Gray conversion: `BGR2GRAY`
- Threshold value, binarization: `threshold`, `THRESH_OTSU`
- Histogram: `hist`

Python/OpenCV - Task 2

Load a test image of your choice and convert it to grayscale. Then write Python programs (or functions) with which you can perform the following tasks.

a) Histogram equalization with `equalizehist`
b) Filters: `GaussianBlur`, `filter2D`, `Sobel`, `Laplacian`.
c) Comparison of a linear low-pass filter (Gaussian, mean value) with a median filter (`medianBlur`)
d) First apply a linear low-pass filter to the image and then a high-pass filter. What happens if you enlarge the window of the low-pass filter (e.g. to 5x5, 7x7, 15x15)? Compare the result with the result you get when you replace the linear low-pass filter with a median.
e) Erosion, dilatation: `erode`, `dilate`
First, write “opening” and “closing” functions based on erosion/dilation. Also compare with the `morphologyEx` function. Special test images are available for this in the Learning Campus.